Problem Set 9: Fuel Cycles & Material Limits

Due: Friday, April 10, 2026

Instructions

- Show all work, including the equations used and unit conversions.
- **Constants & Data:**
  - **Energy Content:** Fissioning 1 gram of any actinide yields ≈ 1 MWd of thermal energy.
  - **Natural Uranium:** Contains 0.7% U-235 and 99.3% U-238.
  - **Zircaloy Cladding:**
    - Mean Radius $r = 4.75$ mm.
    - Wall Thickness $t = 0.6$ mm.
    - Yield Strength $\sigma_y \approx 350$ MPa (at temperature).
  - **Fluence Data:** A typical PWR vessel beltline accumulates a fast fluence of $\Phi_{fast} \approx 2 \times 10^{19}$ n/cm^2 per effective full power year (EFPY).

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1. **The Value of "Waste" (Breeding Physics)**
   In Lecture 27, we discussed how Light Water Reactors (LWRs) utilize only a fraction of the potential energy in natural uranium, while Fast Breeder Reactors (FBRs) can utilize nearly all of it.
   (a) Consider $1,000$ kg of Natural Uranium. In a standard "Once-Through" LWR cycle, we only fission approximately 0.6% of the total mass (mostly the U-235). Calculate the total thermal energy extracted in **GWd** (Gigawatt-days).
   (b) Now consider a closed fuel cycle with Fast Breeder Reactors (FBRs) that can convert and fission the U-238. Assume we can fission 60% of the total mass before losses become prohibitive. Calculate the total thermal energy extracted in **GWd**.
   (c) The U.S. currently stores approximately $700,000$ metric tons of Depleted Uranium (tails). Using the result from (b), estimate the total electrical energy (in TWh) contained in this "waste" stockpile if used in breeders. Assume a thermal-to-electric efficiency of 40%.
   (d) *Context:* The entire U.S. electric grid consumes $\approx 4,000$ TWh per year. How many years could this stockpile power the entire country?

2. **Burnup and Cycle Length**
   A utility operates a 1,200 MWe (electric) PWR with a thermal efficiency of $\eta = 33\%$.
   (a) Calculate the Reactor Thermal Power (P_{th}) in MW.

(b) The core contains 90 Metric Tons of Uranium (MTU). The utility wants to run on an **18-month cycle** (540 days) with a capacity factor of 95% (i.e., the plant runs at full power for 95% of the time). Calculate the required **Cycle Burnup** in MWd/MTU.

(c) If the maximum regulatory limit for fuel rod burnup is $62,000$ MWd/MTU, how many "batches" of fuel must be in the core? (Hint: The discharge burnup is roughly $N \times$ Cycle Burnup, where N is the number of batches). Can they do this with a standard 3-batch core?

3. Atomic Displacements

We measure material damage in **dpa** (displacements per atom).

(a) Consider a fast neutron striking an iron atom in the pressure vessel. The scattering cross-section is $\sigma_s \approx 3$ barns. If the fast flux is $\phi = 10^{11}$ n/cm$^2 \cdot$ s, calculate the **Reaction Rate** (R) (collisions per atom per second).

(b) Each primary collision creates a cascade. Assume the average neutron collision generates $\nu = 500$ secondary atomic displacements. Calculate the rate of displacements (displacements/atom/second).

(c) Over a 60-year operating life (1.9×10^9 seconds), calculate the total **dpa**.

(d) Explain, in physical terms, what your answer in (c) implies about the stability of the crystal lattice. (e.g., "Every atom has been moved...").

4. The DBTT Shift and PTS

A reactor pressure vessel's Ductile-to-Brittle Transition Temperature (DBTT) increases with fluence according to the empirical relationship:

$$\Delta\text{DBTT } (^\circ\text{C}) = 40 + 10 \times \left(\frac{\Phi t}{10^{19}}\right)^{0.5}$$

where Φt is the fast fluence in n/cm^2.

(a) The vessel starts with a fresh DBTT of -20°C. Calculate the DBTT shift and the new absolute DBTT after 40 years of operation (assume fluence rate given in Instructions).

(b) The plant applies for a life extension to 80 years. Calculate the final DBTT at 80 years.

(c) **Safety Check:** During a Small Break LOCA, the Emergency Core Cooling System (ECCS) injects water at 25°C. Is the 80-year-old vessel at risk of Pressurized Thermal Shock (PTS)? Explain why.

5. Cladding Collapse, Operating Pressures, & EOL (Why we pressurize)

A Zircaloy fuel rod is effectively a thin-walled cylindrical pressure vessel.

(a) **Scenario A (Unpressurized):** A fuel rod is manufactured at atmospheric pressure (0.1 MPa). It operates in a reactor where the coolant pressure is 15.5 MPa. Calculate the **Hoop Stress** (σ_θ) on the cladding wall. (Use the thin-wall approximation: $\sigma_\theta = \frac{\Delta P \cdot r}{t}$). Does this put the cladding in tension or compression?

(b) **Scenario B (Pre-pressurized, Cold):** To prevent "Creep Collapse," modern fuel rods are pre-pressurized. Assume the rod is filled with Helium to 3.0 MPa at a manufacturing room temperature of 300 K. Revise your answer for these conditions.

(c) **Scenario C (Pre-pressurized, Hot):** Once the reactor reaches steady-state full power, the heat flux from the fuel pellets raises the volume-averaged temperature of the gas inside the rod to roughly 700 K. Assuming the internal volume of the rod remains constant, calculate the true internal gas pressure during operation and the operating **Hoop Stress** using this hot internal pressure. How much did the magnitude of the stress improve compared to Scenario A?

(d) **Scenario D (End of Life - EOL):** Over the fuel cycle, fission gases (primarily Xenon and Krypton) diffuse out of the fuel pellets and into the rod's gas plenum. By the end of its life, this release has tripled the total number of gas moles (n) inside the rod. Assuming the average gas temperature remains 700 K, calculate the new internal pressure and the final **Hoop Stress**.

(e) Does the EOL stress place the cladding in tension or compression? Briefly explain why fission gas release is a major limiting factor for how long fuel can stay in the reactor.

(f) Why do we initially use Helium specifically, rather than Argon or Nitrogen? (Hint: Think about heat transfer in the "gap" discussed in Lecture 28).